Image Classification with BI Dashboard

Team Gamma

SUMMARY

This project integrates **Deep Learning** and **Business Intelligence (BI)** to create a robust solution for **plant disease detection**. A **Convolutional Neural Network (CNN)** was trained on a plant leaf dataset to classify images into healthy or diseased categories. The model achieved **92.28% training accuracy**, **88.27% validation accuracy**, and **89% test accuracy**, showing strong generalization capability.

To ensure interpretability and stakeholder usability, a **Power BI dashboard** was built. This dashboard visualizes dataset distributions, training progress, model evaluation metrics, and class-level prediction insights. Together, the AI model and BI tool deliver an end-to-end solution for decision-making in agriculture.

INTRODUCTION

Agriculture is one of the most critical sectors, and plant diseases can significantly reduce crop yield and quality. Farmers traditionally rely on manual inspection, which is time-consuming and error-prone. With the advancements in **Artificial Intelligence (AI)** and **Computer Vision**, automated image classification provides a faster, more reliable, and scalable solution.

This project was developed with the following objectives:

1. To train a **deep learning model (CNN)** that accurately classifies plant leaves into healthy and disease categories.
2. To evaluate the model using accuracy, loss, and per-class performance metrics.
3. To design a **Power BI dashboard** that presents results interactively for better decision-making.
4. To provide a real-world system that combines **predictive modeling** and **analytical visualization** for agriculture.

Project Implementation and Results

1. Dataset and Preprocessing

- The dataset contained images of **different plant species** including pepper, potato, and tomato, each with multiple disease categories and healthy classes.
- Total samples used for training, validation, and testing were **over 4,000 images**.
- **Preprocessing steps**:
    - Images resized to a uniform dimension for CNN input.
    - Pixel values normalized (0–1 range).
    - **Data Augmentation** (rotation, flips, zoom) applied to increase dataset variability and reduce overfitting.

- Dataset split into:
  - **Training Set**: 70%
  - **Validation Set**: 20%
  - **Test Set**: 10%

2. CNN Model Architecture

The Convolutional Neural Network (CNN) was built using TensorFlow/Keras.

Layers and Functions:

- **Input Layer**: Accepts preprocessed image tensors.
- **Convolutional Layers (Conv2D)**: Extract local features such as edges, textures, and color patterns.
- **Pooling Layers (MaxPooling2D)**: Reduce spatial dimensions while preserving important features.
- **Dropout Layer**: Prevents overfitting by randomly deactivating neurons during training.
- **Flatten Layer**: Converts feature maps into 1D vectors.
- **Fully Connected Dense Layers**: Combine features and learn non-linear relationships.
- **Output Layer (Softmax)**: Provides probability distribution across multiple classes.

Optimizer & Loss Function:

- Optimizer: **Adam**
- Loss Function: **Categorical Cross-Entropy**
- Evaluation Metrics: **Accuracy, Precision, Recall, F1-score**

3. Model Training and Evaluation

The CNN was trained for **10 epochs**.

Performance Over Epochs:

- **Epoch 1**: Accuracy 44.04% → Val Accuracy 71.65%
- **Epoch 5**: Accuracy 88.38% → Val Accuracy 76.17%
- **Epoch 10 (Final)**: Accuracy **92.28%** → Val Accuracy **88.27%**

Final Results:

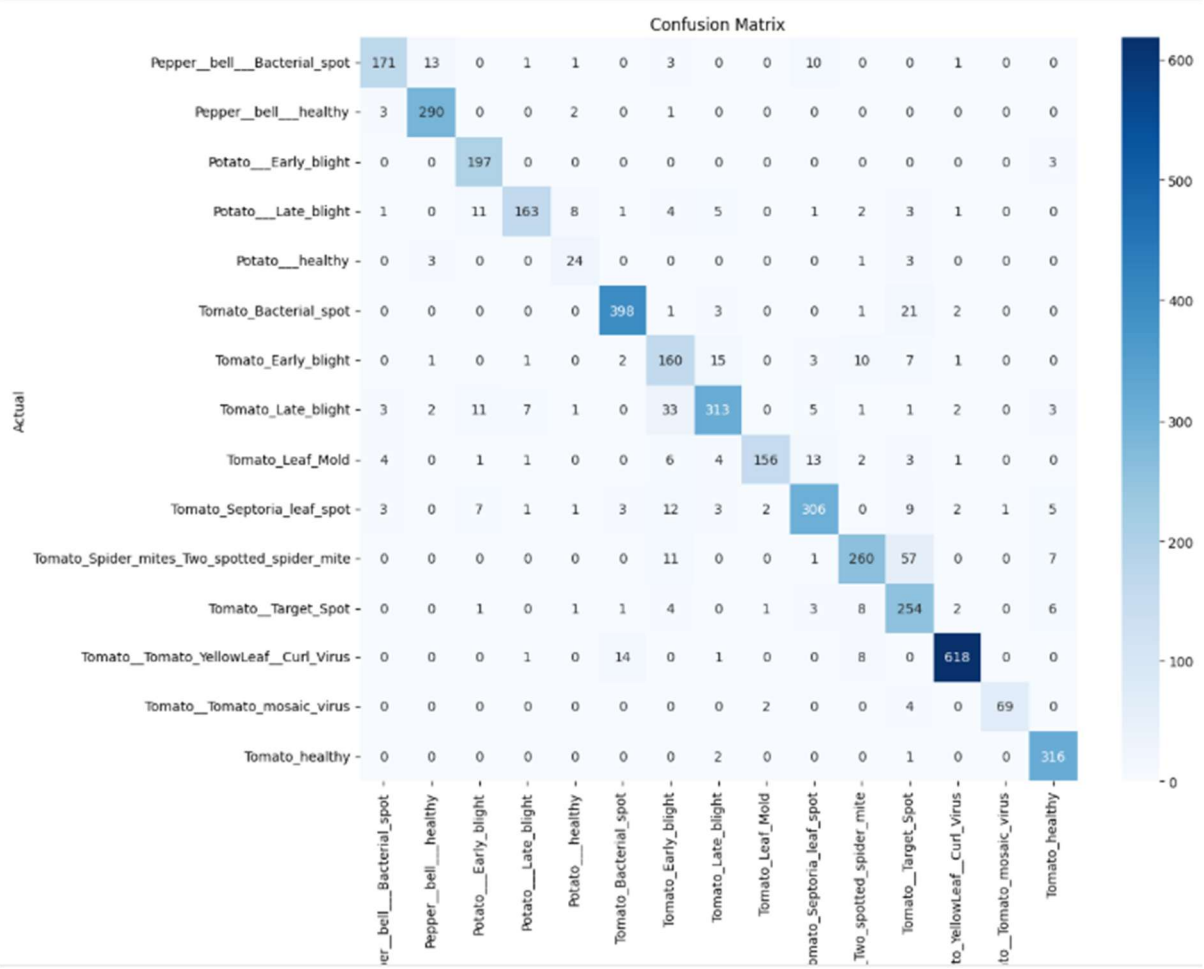
- Training Accuracy: **92.28%**
- Validation Accuracy: **88.27%**
- Training Loss: **0.2201**

- Validation Loss: **0.4131**
- Test Accuracy: **89%**

Classification Report (Highlights):

- *Tomato_YellowLeaf_Curl_Virus*: Precision 0.98, Recall 0.96, F1 0.97
- *Pepper Bell Healthy*: Precision 0.94, Recall 0.98, F1 0.96
- *Potato Healthy*: Precision 0.63, Recall 0.77, F1 0.70 (weaker due to fewer samples)
- Weighted Average F1-score: **0.89**

This shows the model performed strongly on most classes, but classes with fewer samples had reduced accuracy.



The confusion matrix was used to evaluate the model by comparing actual class labels with predicted ones. A heatmap visualization was created, where diagonal cells indicate correct classifications and off-diagonal cells represent misclassifications. This provides insights into the overall accuracy, highlights common errors, and reveals the model's strengths and weaknesses

4. Power BI Dashboard

A **Power BI dashboard** was created using the dashboard.pbix file. It provides an interactive and business-friendly view of model results.

Dashboard Features:

- **Dataset Overview**: Number of images per class and distribution of healthy vs diseased samples.
- **Training Metrics**: Accuracy and loss trends across epochs.
- **Model Evaluation**: Overall accuracy, precision, recall, and F1-score visualized in charts.
- **Confusion Matrix Visualization**: Shows correctly vs incorrectly classified samples.
- **Class-Level Insights**: Drill-down feature to analyze predictions for specific plant diseases.

This dashboard allows stakeholders (e.g., farmers, agronomists, policymakers) to **interpret model outputs easily** and **make informed decisions**.

CONCLUSION

The project successfully demonstrates how **Machine Learning (ML)** and **Business Intelligence (BI)** can be combined for impactful agricultural solutions.

- The CNN achieved **high accuracy (92% training, 88% validation, 89% testing)** in classifying plant diseases.
- **Power BI** enhanced interpretability by presenting results visually and interactively.
- Strong per-class performance on most disease categories validates the effectiveness of the model.
- Weaker accuracy in some minority classes highlights the need for **data balancing** or **further augmentation**.

Future Work:

- Expand dataset with more balanced samples across classes.
- Deploy the model as a **mobile/web application** for farmers.
- Integrate **real-time image capture** for on-field disease detection.
- Extend BI dashboards with **geographic insights** (disease hotspots by region).

In conclusion, the **Image Classification with BI Dashboard** project bridges the gap between **technical AI solutions** and **real-world usability**, making it a scalable and practical tool for smart agriculture.